ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-4 **WINTER SEMESTER, 2024-2025**

DURATION: 40 MINUTES **FULL MARKS: 20**

CSE 4309: Theory of Computing

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. State and prove the pumping lemma for Context-Free Languages (CFLs). $3+5$ (CO2) (PO1)

Solution:

If A is a context-free language, then there exists a number p (the *pumping length*) such that if s is any string in A with $|s| \geq p$, then s may be divided into five pieces

$$s = uvxyz$$

satisfying the following conditions:

1. For each $i \geq 0$,
$$uv^ixy^iz \in A,$$
2. $|vy| > 0$, and
3. $|vxy| \leq p$.

Proof. Let $G = (V, \Sigma, R, S)$ be a CFG generating A. Let

$$b = \max\{\, |\alpha| : (A \to \alpha) \in R \,\}$$

be the maximum branching factor of any node in a parse tree. Define the pumping length

$$p = b^{|V|+1}.$$

Let $s \in A$ with $|s| \geq p$, and choose a parse tree T for s having the *smallest possible number of nodes.*

Finding the repeated variable and the decomposition.

Let H be the height of T and consider a longest root-to-leaf path. Among the last $|V|+1$ variables on this path, some variable must repeat. Let that repeated variable be R.

Thus along the path we have two occurrences of R: an *upper* R and a *lower* R that is a descendant of the upper one (as illustrated in Figure 2).

Because each subtree yields a contiguous substring of s:

$$\text{yield of the upper } R = vxy$$

$$\text{yield of the lower } R = x$$

Hence the whole string has the form

$$s = u(vxy)z$$

so

$$s = uvxyz.$$

We now prove each required criterion using the requested ideas.

(1) Proof that $uv^ixy^iz \in A$ for all $i \geq 0$ (cut–paste argument).

The portion of the tree between the two occurrences of R shows

$$R \Rightarrow^* vRy.$$

The lower subtree shows

$$R \Rightarrow^* x.$$

Therefore we may repeat the derivation $R \Rightarrow^* vRy$ any number of times before applying $R \Rightarrow^* x$:

$$R \Rightarrow^* vRy \Rightarrow^* v(vRy)y \Rightarrow^* \cdots \Rightarrow^* v^iRy^i \Rightarrow^* v^ixy^i.$$

Replacing the subtree at the upper R with this expanded derivation yields a valid parse tree whose yield is

$$uv^ixy^iz.$$

Thus $uv^ixy^iz \in A$ for all $i \geq 0$.

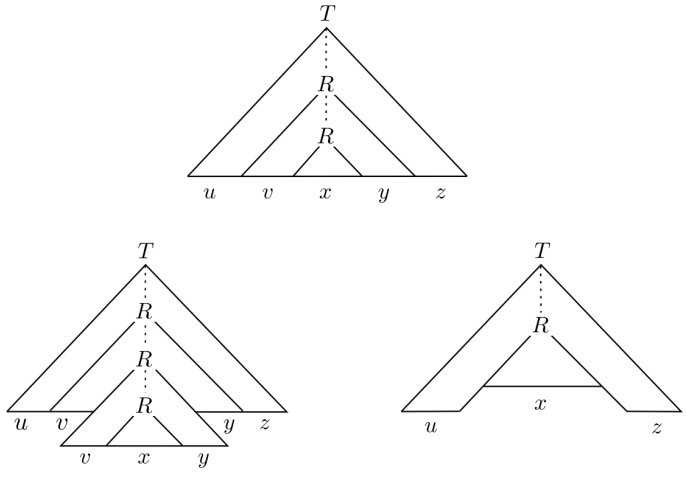

Figure 1: Surgery on parse trees.

(2) Proof that $|vy| > 0$.

Suppose $|vy| = 0$. Then $v = \varepsilon$ and $y = \varepsilon$.

Hence the subtree rooted at the upper R produces exactly the same string as the subtree rooted at the lower R, namely x.

We could therefore replace the entire upper R subtree with the lower one, producing a smaller parse tree that still generates s.

This contradicts the assumption that T has the smallest number of nodes. Hence $|vy| > 0$.

(3) Proof that $|vxy| \leq p$.

Because the upper R occurs among the last $|V| + 1$ variables on a longest root-to-leaf path, the height of the subtree rooted at this R is at most $|V| + 1$.

Since each node has at most b children, a subtree of height h has at most b^h leaves.

Each leaf contributes at most one terminal symbol, so

$$|vxy| \leq b^{|V|+1} = p.$$

Thus conditions (1), (2), and (3) are satisfied, proving the pumping lemma.

□

2. Prove that a language is Turing-recognizable if and only if some **multitape** Turing machine recognizes it. 4 (CO3) (PO1)

Solution:

Proof. We prove both directions by explicit constructions.

(⇒) Assume L is Turing-recognizable. Then there exists a (single-tape) TM

$$M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{acc}}, q_{\text{rej}})$$

that recognizes L.

Construct a k-tape TM M' (for any fixed $k \geq 1$) that recognizes L as follows: on input w, M' places w on tape 1 (as usual) and simply simulates M step-for-step using only tape 1, ignoring tapes $2, \ldots, k$. Thus M' accepts exactly when M accepts. Hence M' recognizes L.

(⇐) Assume some k-tape TM M recognizes L. We construct a *single-tape* TM S that recognizes L by simulating M.

Let M have tape alphabet Γ and transition function

$$\delta : Q \times \Gamma^k \to Q \times \Gamma^k \times \{L, R\}^k.$$

(If δ is partial, undefined transitions may be treated as immediate reject/loop; this does not affect recognizability.)

Encoding of k tapes on one tape. Machine S uses one tape to store the contents of all k tapes of M in blocks separated by a special delimiter symbol $\#$ (where $\# \notin \Gamma$). To record head positions, S uses a marked ("dotted") version of each tape symbol:

$$\dot{\Gamma} = \{\dot{a} : a \in \Gamma\}, \qquad \Gamma' = \Gamma \cup \dot{\Gamma} \cup \{\#\}.$$

At any time, in each block exactly one symbol is dotted, indicating the head position of that simulated tape.

On input $w = a_1 \cdots a_n$, S writes the initial encoding

$$\# \dot{a}_1 a_2 \cdots a_n \# \dot{\sqcup} \# \dot{\sqcup} \# \cdots \# \dot{\sqcup} \#$$

(i.e., tape 1 contains w with its head at the first symbol, and tapes $2, \ldots, k$ are blank with heads at the leftmost blank).

Simulating one step of M. Suppose the current simulated state is $q \in Q$.

1. S scans its tape left-to-right to locate the k dotted symbols (one in each block) and records in its finite control the underlying symbols $a_1, \ldots, a_k \in \Gamma$ currently under the simulated heads.

2. S computes (in its finite control) the transition of M:
$$\delta(q, a_1, \ldots, a_k) = (q', b_1, \ldots, b_k, d_1, \ldots, d_k),$$
where each $b_i \in \Gamma$ is the symbol to be written and each $d_i \in \{L, R\}$ is the head movement for tape i.

3. S makes another scan to update each block: it replaces the dotted symbol $\dot{a}_i$ by $\dot{b}_i$ *temporarily*, and then moves the dot one position left or right according to d_i:

   - If $d_i = R$, S undots the current cell (leaving b_i) and dots the symbol immediately to the right; if the head is at the right end of the block, S first inserts a blank symbol before the next delimiter so that the move is possible.
   - If $d_i = L$, S undots the current cell (leaving b_i) and dots the symbol immediately to the left; similarly, if needed it ensures a blank exists at the left end of the block.

   Finally, S updates its simulated state to q'.

This completes the simulation of one transition of M using finitely many steps of S.

Correctness (recognition is preserved). By construction, after t simulated steps, the single tape of S encodes exactly the k tapes of M after t steps, including head positions and current state. Therefore:

- If M ever enters q_{acc} on input w, then S reaches the corresponding simulated configuration and enters q_{acc} as well, so S accepts w.
- If M does not accept w (i.e., it rejects or loops), then S likewise never accepts w (because it faithfully simulates M step-by-step).

Hence S recognizes exactly the same language as M, namely L.

Since we have constructed a single-tape recognizer from any multitape recognizer, L is Turing-recognizable. □

3. What is the time and space complexity of the Cocke–Younger–Kasami (CYK) algorithm? 3 (CO1) (PO1)

Solution:

Let w be an input string of length n, and let G be a context-free grammar in Chomsky Normal Form (CNF) with $|V|$ variables and $|P|$ productions.

- **Time complexity:** The CYK algorithm runs in
$$O(n^3 \cdot |P|),$$

often written as $O(n^3 \cdot |G|)$ depending on how the grammar size is measured. (Equivalently, if one bounds by the number of variables, it is sometimes stated as $O(n^3 \cdot |V|^3)$ in the worst case.)

- **Space complexity:** The CYK table has $\Theta(n^2)$ cells, and each cell can store up to $|V|$ variables, so the space usage is
$$O(n^2 \cdot |V|),$$
which is often reported simply as $O(n^2)$ if G is treated as fixed.

4. An internet meme is defined as a humorous or culturally relevant image, video, or text that spreads rapidly online, often evolving as it is shared and adapted by users. Draw a rough sketch of a meme related to this course. 5 (CO69) (PO420)
Warning: Cringe and lame memes will result in negative marking.

Solution:

Figure 2: Cortisol ↑↑.

Rubric:

- 5 points for a good meme.